PRAKTIKUM SISTEM OPERASI

MODUL 10

SHELL

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.
   Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.
   a. Pada prototypes.h chname diubah menjadi:

```
extern  status  ascdate(uint32, char *);

/* in file bufinit.c */
extern  status  bufinit(void);

/* in file chprio.c */
extern  syscall chprio(pid32, pri16);

/* in file chname.c */
extern  pri16   chname(pid32, char *, uint32);

/* in file clkupdate.S */
extern  uint32  clkcount(void);
```

   b. Pada chname.c fungsi diubah dari:

```
/*------------------------------------------------------------------------
 *  chname  -  Dummy syscall
 *------------------------------------------------------------------------
 */
syscall chname(
        pid32           pid,
        char *          newname
)
```

   menjadi

```
#include <xinu.h>

/*------------------------------------------------------------------------
 *  chname  -  Dummy syscall
 *------------------------------------------------------------------------
 */
pri16 chname(
        pid32           pid,
        char            *newname,
        uint32          namelen
)
```

   Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil

```
 *------------------------------------------------------------------------
 */
pri16 chname(
        pid32           pid,
        char            *newname,
        uint32          namelen
)
{
        intmask mask;
        struct procent *prptr;
        uint32 i;

        mask = disable();

        /* Validasi PID*/
        if (isbadpid(pid)) {
                restore(mask);
                return SYSERR;
        }

        prptr = &proctab[pid];

        /* Nama baru*/
        for (i = 0; i < namelen && i < PNMLEN-1; i++) {
                prptr->prname[i] = newname[i];
        }

        /* Null terminator */
        prptr->prname[i] = '\0';

        restore(mask);
        return OK;
}
```

2. Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

   a. Edit shell.c

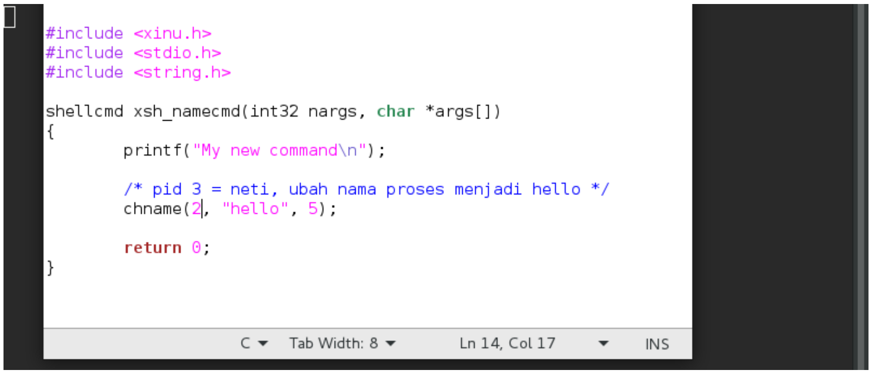
```
#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/************************************************************************/
/* Table of Xinu shell commands and the function associated with each   */
/************************************************************************/
const   struct  cmdent  cmdtab[] = {
        {"argecho",     xsh_argecho},
        {"arp",         xsh_arp},
        {"cat",         xsh_cat},
        {"clear",       xsh_clear},
        {"date",        xsh_date},
        {"devdump",     xsh_devdump},
        {"echo",        xsh_echo},
        {"help",        xsh_help},
        {"ls"   ,       xsh_ls},
        {"kill",        xsh_kill},
        {"memdump",     xsh_memdump},
        {"memstat",     xsh_memstat},
        {"ns",          xsh_ns},
        {"netinfo",     xsh_netinfo},
        {"ping",        xsh_ping},
        {"ps",          xsh_ps},
        {"sleep",       xsh_sleep},
        {"tee",         xsh_tee},
        {"udp",         xsh_udpdump},
        {"udpecho",     xsh_udpecho},
        {"udpeserver",  xsh_udpeserver},
        {"uptime",      xsh_uptime},
        {"namecmd",     xsh_namecmd},
        {"mycmd",       xsh_mycmd},
        {"?",           xsh_help}
```

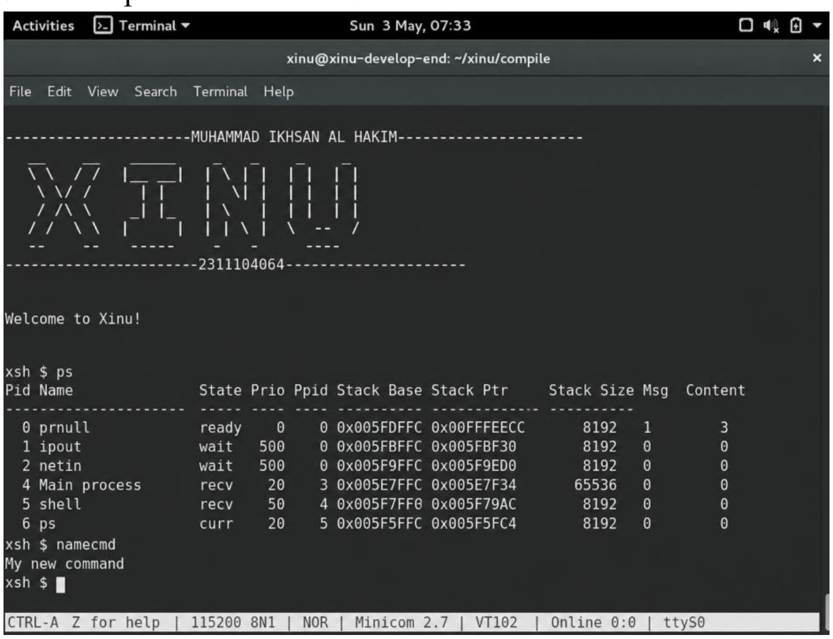
   b. Edit shprototypes.h

```
/* in file xsh_namecmd.c */
extern  shellcmd  xsh_namecmd   (int32, char *[]);
```

   c. Buat file baru:



3. Test hasilnya:
   a. Masuk ke terminal xinu
   b. Jalankan perintah ps
   c. Jalankan perintah namecmd



d. Jalankan perintah ps

```
Welcome to Xinu!


xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFECC     8192   1        3
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192   0        0
  2 netin            wait   500    0 0x005F9FFC 0x005F9ED0     8192   0        0
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536   0        0
  5 shell            recv    50    4 0x005F7FFC 0x005F79AC     8192   0        0
  6 ps               curr    20    5 0x005F5FFC 0x005F5FC4     8192   0        0
xsh $ namecmd
My new command
xsh $ ps
Pid Name             State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
--- ---------------- ----- ---- ---- ---------- ---------- ----------
  0 prnull           ready    0    0 0x005FDFFC 0x00FFFED8     8192   1        3
  1 ipout            wait   500    0 0x005FBFFC 0x005FBF30     8192   0        0
  2 hello            wait   500    0 0x005F9FFC 0x005F9ED0     8192   0        0
  4 Main process     recv    20    3 0x005E7FFC 0x005E7F34    65536   0        0
  5 shell            recv    50    4 0x005F7FFC 0x005F79AC     8192   0        7
  8 ps               curr    20    5 0x005F5FFC 0x005F5FC4     8192   0        0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

e. Lihat nama proses telah berubah
Nama proses berubah dari *netin* menjadi *hello* karena syscall chname() berhasil dijalankan melalui command namecmd.